

Index Optimization: Learned Index Structures Replacing Inverted Indexes for Large-Scale Semantic Retrieval

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Abstract—Inverted Indexes are excellent for exact-match keyword searches, they are unsuitable for modern vector-based semantic search due to the "Curse of Dimensionality." In this paper, we consider exploring graph-based indexing algorithms—such as HNSW (Hierarchical Navigable Small World)—or Product Quantization (PQ) for vector compression and approximate search. Additionally, consider the "Learned Index" approach, which employs machine learning to predict storage locations rather than relying on traditional B-Trees.

GitHub link: https://github.com/SalvadorDalier/TEAM_10_SEG.git

I. INTRODUCTION

Inverted index has been a great solution in looking for keywords and its exact match in search engines. However, it suffers from Curse of Dimensionality, therefore becomes unreliable in searching based on context rather than word (vector search). In this paper, we focus on researching graph index algorithms (such as HNSW - Hierarchical Navigable Small World) or Product Quantization (PQ) for compression and approximate search on vectors and proposing a "Learned Index" model that uses machine learning to predict storage locations, instead of using traditional B-Tree.

II. RELATED WORK

This research sits at the intersection of Inverted Indexing, Approximate Nearest Neighbor (ANN) search, and Machine Learning-based Indexing Structures.

Traditional Information Retrieval systems (such as Apache Lucene and Elasticsearch) rely on Inverted Indexes for exact keyword matching. However, when applied to dense vector embeddings, they suffer from the "Curse of Dimensionality," resulting in exponential index growth, high memory overhead, and severe query latency.

To achieve scalable vector search, ANN algorithms trade marginal accuracy for substantial throughput gains. The Hierarchical Navigable Small World (HNSW) graph by Malkov and Yashunin (2018) [1] provides state-of-the-art query speeds via multi-layer routing, though at the expense of high RAM usage. To resolve memory bottlenecks, Product Quantization (PQ) by Jégou et al. (2010) [2] compresses vector spaces into centroid codes, drastically lowering memory footprint while maintaining fast distance lookups.

Concurrently, Kraska et al. (2018) [3] introduced Learned Index Structures, replacing static tree traversals with ML models that predict data locations directly via Cumulative Distribution Functions. Slanináková et al. (2022) [4] extended this concept to metric spaces with the Learned Metric Index (LMI).

This study combines these paradigms by deploying a Learned Index to predict optimal routing entry points within an HNSW and PQ pipeline, offering a memory-efficient, high-throughput replacement for traditional Inverted Indexes in large-scale semantic search.

III. METHODOLOGY

The proposed vector database system architecture is designed based on a combination of three core components—Learned Index, HNSW Graph, and Product Quantization (PQ) — to replace traditional B-Tree or Inverted Index structures. The system is divided into two main phases:

A. Indexing Phase (Initial Data Processing)

- Data Compression: Data is broken down into smaller segments and replaced with representative codes to conserve memory.
- Link Network Creation: The compressed data is interconnected to form a multi-layer network, positioning similar data points close to one another.
- Position Model Training: The system employs a simple machine learning model to learn data organization, enabling it to rapidly predict the storage location of specific data items rather than relying on manual searching.

B. Query Phase (Search)

- The user's search request is converted into a compact, lightweight format.
- The machine learning model quickly predicts the most likely storage region.
- The system navigates the link network within that region to select results that are semantically closest to the request.

IV. EXPERIMENTS

This section presents the experimental setup, evaluation metrics, and comparative analysis of three indexing paradigms:

the traditional Inverted Index, HNSW, and the proposed Learned Index.

A. Experimental Setup

Environment and Dataset: All algorithms were implemented and benchmarked under identical hardware configurations to ensure a fair comparison. The evaluation dataset comprises high-dimensional semantic vector embeddings queried under real-world retrieval workloads. **Compared Baselines:** Inverted Index (Baseline): A traditional sparse term/posting-list index structure. HNSW: A multi-layer graph-based Approximate Nearest Neighbor (ANN) search algorithm. Learned Index: An ML-driven routing approach that predicts entry points within a combined HNSW and Product Quantization (PQ) pipeline.

B. Evaluation Metrics

System performance is evaluated across three primary metrics: + Recall@3: The proportion of ground-truth top-3 items successfully retrieved (measuring search accuracy). + QPS (Queries Per Second): The number of search queries processed per second (measuring throughput and speed). + Memory Consumption (MB): The total RAM occupied by the index structure (measuring memory efficiency).

C. Experimental Results

The benchmark results comparing the three methods are summarized below:

TABLE I
BENCHMARK PERFORMANCE COMPARISON ACROSS INDEXING METHODS

Method	Recall@3	QPS	Memory (MB)
Inverted	0.7667	4,223.00	591.64
HNSW	1.0000	189,846.26	591.75
Learned	1.0000	12,686.78	587.14

V. DISCUSSION

This section presents a deeper analysis of the technical trade-offs, the underlying factors influencing performance variations across architectures, and the practical viability of the Learned Index model.

A. Accuracy-Speed-Memory Trade-offs

The experimental findings reveal clear trade-offs across the three indexing paradigms:

- **Inverted Index vs. Vector Approaches:** While inverted indexes are traditionally regarded as memory-efficient structures for sparse text data, empirical results indicate that they suffer severe degradation when applied to continuous semantic vector embeddings, achieving a low Recall@3 of 0.7667 and the lowest throughput at 4,223.00 QPS. This confirms that inverted posting lists are fundamentally ill-suited for high-dimensional semantic search workloads.
- **HNSW vs. Learned Index:** Standalone HNSW delivers exceptionally high search throughput (189,846.26 QPS)

due to direct multi-layer graph navigation in RAM. However, the Learned Index achieves a strategic balance: it maintains perfect accuracy (Recall@3 = 1.0000), delivers a 3× speedup over the Inverted Index baseline, and records the lowest memory footprint among all three evaluated methods (587.14 MB).

B. ML Inference Overhead Analysis

The search throughput of the Learned Index (12,686.78 QPS) is lower than that of pure HNSW. The primary cause is **predictive model inference overhead**:

- 1) Upon receiving a query, the Learned Index model must perform forward-pass calculations to estimate the Cumulative Distribution Function (CDF) or data space partitioning in order to identify optimal search entry points.
- 2) This predictive step introduces a slight latency penalty before execution advances to the local candidate distance search phase.

Despite this inference overhead, accurate predictive routing effectively narrows down the candidate search space, preserving perfect retrieval quality (Recall@3 = 1.0000).

C. Scalability and Index Compression Potential

Although all three evaluated architectures exhibit similar memory footprints (~587 MB to 591 MB) at the current scale, the Learned Index saved approximately 4.61 MB relative to standalone HNSW. This property becomes increasingly significant as dataset size scales to millions or billions of vectors:

- Graph-based HNSW indices suffer from memory growth proportional to the total number of inter-node edges, creating RAM capacity bottlenecks on massive collections.
- Integrating Product Quantization (PQ) alongside predictive modeling in the Learned Index compresses high-dimensional vectors into compact centroid codes. As the dataset expands, the model parameter size remains virtually constant, allowing total index memory usage to grow at a substantially lower rate than pure HNSW graphs.

D. Limitations and Future Directions

- **Limitations:** The current evaluation focuses primarily on read-heavy, static retrieval workloads. The operational overhead of model retraining during continuous data insertions, updates, and deletions was not evaluated in detail.
- **Future Work:** Subsequent research will focus on developing dynamic online learning techniques for continuous index updates and leveraging hardware acceleration (GPU/NPU) during the ML inference step to narrow the throughput gap with standalone HNSW graphs.

VI. CONCLUSION

This research evaluated the replacement of traditional Inverted Indexes with advanced vector architectures—specifically standard HNSW and a Learned Index approach—across accuracy, search throughput, and memory consumption. The experimental results demonstrate that vector-based methods significantly outperform the traditional Inverted Index in both retrieval quality and execution speed. While the Inverted Index achieved a Recall@3 of 0.7667 at 4,223.00 QPS, both HNSW and the Learned Index reached perfect retrieval accuracy (Recall@3 of 1.0000). In terms of query throughput, standalone HNSW delivered the highest search speed at 189,846.26 QPS, whereas the Learned Index achieved 12,686.78 QPS—representing a $3\times$ speedup over the baseline Inverted Index while matching the accuracy of HNSW. Memory consumption across all three architectures remained nearly identical (587 MB to 591 MB), indicating that transitioning to ML-driven and graph-based vector structures introduces virtually no RAM overhead while providing substantial gains in search accuracy and throughput. Consequently, adopting learned and graph-based vector database techniques offers a superior, highly efficient alternative to traditional inverted indexing for modern semantic search systems.

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